

Jianfeng Zhang

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EDUCATION

- **Wuhan University** Wuhan, China
Bachelor of Science in Applied Mathematics; GPA: 3.64; Ranking: 3/42 Sep. 2014 – Jun. 2018

RESEARCH EXPERIENCE

- **Image Segmentation Algorithm Based on Alternating Descent Solver** Wuhan University, China
Research Assistant Mar. 2018 - Present
 - Built a Mumford-Shah-like model framework for interactive image segmentation.
 - Applied geodesic distance in LAB space to build unary term and utilized L1, L2 total variation, and fast-bilateral-solver-like regularization as pairwise term.
 - Developed an approach to optimize unary and pairwise terms alternately to perform image segmentation. This algorithm can reach 0.8 mean IoU in graphcut dataset with little interaction.
- **Long Short-Term Memory (LSTM) for Water Table Depth Prediction** Wuhan University, China
Research Assistant Apr. 2017 - Jan. 2018
 - Developed a LSTM-based model to forecast water stable depth in Hetao Irrigation District. The final best model achieved 0.93 R^2 score on prediction tasks.
 - Built a GUI to construct a decision system based on this model, to aid engineers and decision-makers to plan and manage groundwater resources in Hetao Irrigation District.
 - Our research was published in **Journal of Hydrology**, and I served as first author.[\[Paper\]](#) [\[Code\]](#)

WORK EXPERIENCE

- **Farsee2 Inc.** Wuhan, China
Research Intern with Dr. Xiaoping Zhang Nov. 2017 - Present
 - Built a video object segmentation framework based on CNN and user interaction. this framework is currently embedded in company products.
 - Reviewed current semantic and instance segmentation models and built a model based on deformable convolutional network and ConvCRF for instance segmentation.
- **SinoCLC Inc.** Beijing, China
Data Mining Intern Jul. 2017 - Sep. 2017
 - Developed a web crawler to collect daily text data from Weibo and News Websites using Python with Scrapy.
 - Built a text classification system using CNN, RNN and Word Embeddings to help data analysts find news of interest efficiently.

PROJECTS

- **ICPR MTWI 2018 Challenge on Tianchi** Mar. 2018 - May. 2018
 - Built a faster-rcnn like model for end to end network image text detection and recognition. This model Ranked 13/1488 on [Tianchi Challenge](#).
- **Machine Learning Seminar** Oct. 2017 - Present
 - Organized weekly machine learning seminar in Wuhan University, including professors and students.
 - Gave talks about convex optimization, deep learning and their applications.

TEACHING EXPERIENCE

- **C Programming Language** Spring 2017
- **Data Structures and Algorithms in Python** Fall 2017

PROGRAMMING SKILLS

- **Languages:** Python, MATLAB, C/C++, R
- **Deep Learning Tools:** PyTorch, Tensorflow, Keras